25901592

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZIM201-MEDICAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks:50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks
Answer all questions
Each question carries 1 marks

1. The gut flora helps in…… of digested carbohydrates.[K]/[C01]
Answers :fermentation
2. Actinomyces is a gram positive ……. bacterium [U]/[CO4]
Answer: Anaerobic
3. Severe dengue is also called …….. [K]/[C05]
Answer:Dengue Hemorrhagic Fever
4. The normal microflora present in the digestive system is known as the…….. [K]/[C01]
Answer: Gut microbiota
5. Prolonged ……predisposes to candidasis [K]/[CO5]
Answer: Antibiotic therapy
6. A carrier who has recovered from a disease but still spread pathogens is ……
Answer:Convalescent Carrier **[K]/[CO2]**
7.The type of paralysis caused by letanus is …….. [K]/[CO4]
Answer:Spastic paralysis
8. Hepatitis B virus contains……. as its genetic material.[U]/[C05]
Answer: Partially double stranded DNA
9…….. commonly causes fungal genital infections [K]/[CO3]
Answer: Candida albicans
10. Streptococcus pyogenes commonly causes …….. in the respiratory tract. [K]/[CO4]
Answer:Pharyngitis **[1x10= 10]**

Part B

Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

11. Define amoebiasis. [A]/[CO5]
Amoebiasis is a parasitic infection of the intestines caused by the protozoan Entamoeba histolytica. It is characterized by symptoms ranging from mild diarrhea to severe dysentery with blood and mucus in the stool

12. Describe the causative agent of diptheria. [K]/[CO4]

The causative agent is Corynebacterium diphtheriae. It is a gram-positive, non-motile, club-shaped bacillus that typically arranges in "Chinese letter" patterns. It produces a potent exotoxin that causes the characteristic pseudomembrane in the throat.

13. What is swine flu? Mention its causative agent. [A]/[CO5]

Swine flu is a respiratory disease caused by type A influenza viruses that regularly cause outbreaks in pigs. The most well-known causative agent affecting humans is the H1N1 virus strain.

14. What is invasion? [K]/[CO2]

Invasion refers to the process by which microorganisms enter host cells or tissues, spread, and establish an infection. It often involves the secretion of enzymes or proteins that break down host physical barriers.

15. Explain the pathogenicity and complications of brucellosis.[U]/[CO4]

Pathogenicity: Brucella species are intracellular pathogens that survive within phagocytes, spreading through the lymphatic system to the liver, spleen, and bone marrow. It typically causes "undulant fever."

Complications: Can include endocarditis, osteomyelitis (bone infection), meningitis, and chronic fatigue.

16. Define colonization resistance. [K]/[C01]

Colonization resistance is the mechanism by which the normal microflora protects the host against the entry and overgrowth of exogenous pathogens by competing for nutrients and attachment sites.

17. Name two organs affected by extra pulmonary tuberculosis. [U]/[CO4]

Extrapulmonary TB can affect organs outside the lungs, most commonly:

Lymph nodes (Scrofula)

Kidneys (Renal TB)

(Other examples include the spine/Pott's disease and the brain/meningitis).

18. Define syphilis. [A]/[CO5]

Syphilis is a chronic, systemic sexually transmitted infection (STI) caused by the spirochete bacterium Treponema pallidum. It progresses through primary, secondary, latent, and tertiary stages if left untreated.

19. Define leprosy and name the caustive organism. [K]/[CO4]

Leprosy (Hansen's disease) is a chronic infectious disease that primarily affects the skin, peripheral nerves, and mucosal surfaces of the upper respiratory tract. The causative organism is Mycobacterium leprae.

20. How is pertussis transmitted? [U]/[CO4]

Pertussis (Whooping Cough) is highly contagious and is transmitted through respiratory droplets produced when an infected person coughs or sneezes.

21. Define upper respiratory tract infections [K]/[CO3]

URTIs are illnesses caused by an acute infection which involves the upper respiratory tract, including the nose, sinuses, pharynx, or larynx. Common examples include the common cold, tonsillitis, and pharyngitis.

22. Complication of pneumonia. [U]/[CO4]

Common complications include pleural effusion (fluid around the lungs), empyema (pus in the pleural cavity), lung abscesses, and bacteremia (which can lead to septic shock).

[2x10 = 20]

Part C
Short Essay Questions
Answer any 5 question
Each question carries 4 marks

23. Describe the role of antifungal drugs in the treatment of cutaneous and systemic mycose.(Cutaneousmycoses-2marks,systemicmycoses-2marks)

[U]/[CO5]

Antifungal drugs disrupt fungal cell membranes by targeting ergosterol or cell wall synthesis. For cutaneous mycoses, topical agents like clotrimazole are used to treat localized infections of the skin and nails. Systemic mycoses require intravenous drugs like Amphotericin B to treat

life-threatening internal infections. Their primary role is to inhibit fungal growth while minimizing damage to host cells

24. Explain carrier state in typhoid fever and describe how carriers contribute to disease transmission.(Definition of carriers-1marks, Mechanism of transmission- 2 marks, Site of persistance-1 marks)
[U] /[CO4]

A carrier is an asymptomatic person who harbors Salmonella typhi in their gallbladder after recovering from typhoid. These individuals intermittently shed the bacteria through their feces and urine into the environment. They contribute to disease transmission by contaminating food and water, often through poor hand hygiene. Identifying and treating carriers is essential for controlling community outbreaks and long-term prevention.

25. Explain in detail the mechanism of action of chloroquine and artemisinin. (Mechanism of action of chloroquine -2 marks, artemisinin-2 marks) [U]/[CO5)

Chloroquine prevents the parasite from converting toxic heme into non-toxic hemozoin within the food vacuole, leading to parasite poisoning. Artemisinin reacts with iron in the heme to generate toxic free radicals that damage the parasite's proteins and membranes. Together, these mechanisms effectively clear Plasmodium parasites from the human bloodstream. Both drugs are vital components in the global fight against malaria

26. Discuss tuberculosis in detail. (Etiology & Transmission-1 Marks, Pathogenesis -1 marks, Clinical features and management -2 marks) [U]/[CO3]

Tuberculosis is a chronic airborne infection caused by Mycobacterium tuberculosis that primarily targets the lungs. It is characterized by the formation of granulomas, where bacteria can remain dormant for long periods. Transmission occurs through respiratory droplets from coughing or sneezing by an infected person. Treatment requires a multi-drug regimen, typically including Isoniazid and Rifampicin, over several months to ensure total eradication

27. Explain the epidemiology virulence factors and pathogenesis of Haemophilus influenzae infections in the respiratory infections.(Epidemology-1marks,Virulence factors -2 marks,Pathogenesis-1marks) [K]/[CO4]

H. influenzae is a common respiratory pathogen whose main virulence factor is a polysaccharide capsule that prevents phagocytosis. It colonizes the nasopharynx and can cause local infections like sinusitis or systemic diseases like meningitis. Epidemiology shows it primarily affects young children, though vaccines like Hib have drastically reduced severe cases. Adhesion factors and IgA proteases also play key roles in its ability to infect human mucosal surfaces.

28. Describe antiretriviral therapy and its role in the management of AIDS.(Definition & Mechanism-2marks, Management - 2 marks) [U]/[C05]

ART uses a combination of drugs to block different stages of the HIV life cycle, such as reverse transcription and integration. This treatment suppresses the viral load to undetectable levels, allowing the immune system to recover and maintain CD4 counts. While ART is not a cure, it prevents the progression to AIDS and significantly reduces the risk of virus transmission. Regular adherence is critical to avoid the development of drug-resistant HIV strains.

29. Discuss malaria as a parasitic disease of global importance including its biological characteristics.(Globalsignificance-2Marks,Lifecycle-2Marks)

[K]/[C05]

Malaria is a major global health threat caused by Plasmodium parasites and transmitted by the female Anopheles mosquito. It features a complex life cycle involving an asexual stage in human liver and blood cells and a sexual stage in mosquitoes. The disease is biologically characterized by recurring fevers and anemia due to the destruction of red blood cells. It remains a significant challenge in tropical regions due to increasing drug and insecticide resistance.

[4x5 = 20]